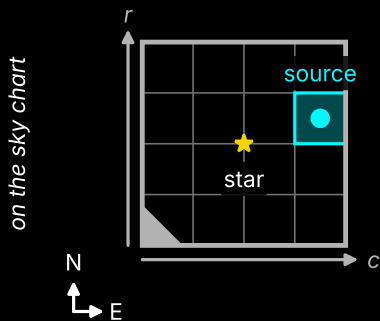
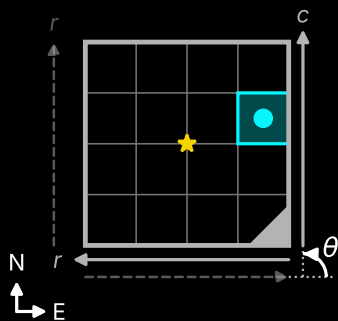
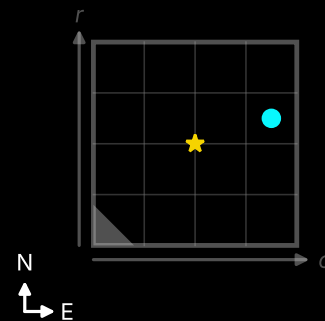
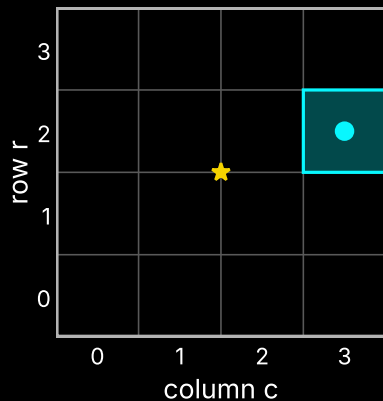
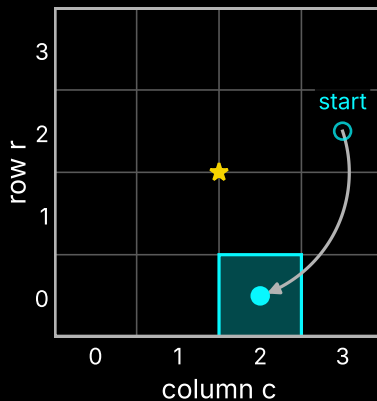
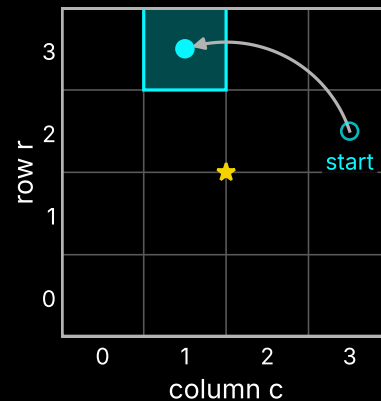


(a) roll 0

(b) roll $\theta = +90^\circ$ (c) active rotation $+90^\circ$ source in $(r, c) = (2, 3)$ source in $(r, c) = (0, 2)$ 

nothing physical turns

 $(r, c) = (2, 3)$
starting pixel $(r, c) = (0, 2)$
sky turns clockwise $(r, c) = (3, 1)$
content turns
counterclockwise

schematic; origin at the grid center; proposed pixel profile; image-coordinate profile pending